may be an activity on the horizon, and there are reasonable concerns about the ability to adequately protect interests in prospective 'mining' sites (after all exclusivity is pretty indistinguishable from appropriation) but the actual activity of prospecting is little different from explorative and scientific activities which are clearly permitted by the Outer Space Treaty and broader body of space law. Further, there is certainly no reason to suppose that an activity that has been conducted by several government space agencies should be proscribed for non-governmental entities.

While predicting timelines for technological developments is difficult, the history of the space industry suggests that one should assume a longer development timeframe, especially when the industry is advancing such a timeframe themselves.[235] It seems reasonable to assume a 20-30-year development timeframe for space resource activities given that there have as yet been no prospecting missions, no mining equipment developed, and the time it takes to get to the objects that these entities wish to mine.[236] Therefore, especially given the demise of the two leading contenders for the status of "pioneer operators"[237] it is clear that there is abundant time to consider an appropriate regime. This is important as there is a need to accommodate several potentially conflicting aspects.

The Moon needs protecting. This does not mean a complete moratorium on space resource activities, or indeed any activity on the Moon but there does need to be a recognition that there are sites of historical, scientific, aesthetic and cultural value that need to be protected. This is potentially true on other planets and moons but given the centrality of the Moon to humanity for millennia it is particularly true on our Moon. Furthermore, this does not necessarily need any "hard law" options, a code of conduct, or some other "soft law" agreement, like the space debris mitigation guidelines, that States, through their national legislation, require their nationals to comply with (i.e. it's binding via national law but not on States via international law). There are at least two efforts underway to do things like this, be it some of the stipulations in The Hague Space

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[235] Mike Wall, *Asteroid Mining May be a Reality by 2025,* SPACE.COM (Aug. 11, 2015), https://www.space.com/30213-asteroid-mining-planetary-resources-2025.html.

[236] Matt Williams, *How Long Does it Take to Get to the Asteroid Belt?* UNIVERSE TODAY (Aug. 10, 2016), https://www.universetoday.com/130231/long-take-get-asteroid-belt/.

[237] HAGUE BUILDING BLOCKS, *supra* note 72, at ¶ 4.2(k).

Resources Governance Working Group's Draft Building Blocks[238] and For All Moonkind's registry of historical sites.[239]

There is also an equity and even sustainability issue to give consideration. As discussed above economically viable asteroids are not necessarily as common as we think. Therefore, there is a responsibility to ensure equitable access to resources for all states. There is a moral imperative to do this but also a legal responsibility given the provisions of Article I of the Outer Space Treaty. Therefore, there is a need to insure a reserve of (comparatively) easily accessible ore bearing objects for those States which will be late to the space mining game owing to historical and developmental handicaps. This is particularly the case if the number of ore bearing objects is towards the lower end of the scale.

It is also important to recognize that it is beneficial to allow property rights to develop from the "grassroots" and that there are potentially negative consequences for imposing a top-down system of property rights, especially when those who are doing the imposing have limited stakes in their development. Not only can it prove to be economically inefficient, especially when a "one size fits all" model is applied (as for example with the US Homestead Act which imposed a uniform model of farm on wildly different climatic areas of North America[240]) but it is important that the actors involved have a stake in the process, if the property rights regime doesn't work it will be circumvented, and this may lead to conflict, which would be precisely what any international space resources activities regime would be designed to prevent.

Additionally, it is vital that property rights are enforceable, granted this should be a given, but it is potentially a bigger issue than it may seem at first. As discussed, part of the issue with the current approach being taken to the regulation of space resource activities is that it is being done at the national level. In and of itself this is not particularly an issue, and States do have an obligation under Article VI of the Outer Space Treaty to "authorize and supervise" the activities of their nationals in outer space and therefore do

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[238] *Id. at* ¶ 17(b).

[239] *See* Roy Balleste & Michelle Hanlon, *How Blockchain Technology can Track Humanity's Lunar Heritage Sites,* THE SPACE REVIEW (June 11, 2018), http://thespacereview.com/article/3512/1.

[240] ANDERSON & HILL, *supra* note 199, at 168-74.

need a domestic legal framework for doing so.[241] However, in order to avoid any issues relating to the provisions of Article II of the Outer Space Treaty[242], the basis for this legislation needs to be on the personal jurisdiction that States enjoy over their nationals. This jurisdiction does not extend to foreign nationals, which could make protecting property rights over space resources particularly problematic without a multilateral framework for some form of mutual recognition (as envisioned by The Hague Working Group's Draft Building Blocks.[243] Again, this is a potential source of conflict, especially if space, and space resources, come to be seen as an increasingly strategic asset.

Furthermore, property rights evolve and adapt to suit novel situations and circumstances. When miners, ranchers and farmers spread into the American West it became clear that the water rights regime that had worked well in water abundant Western Europe and had been successfully transplanted to the Eastern US was not suitable for the arid conditions in the American west, so a new approach was devised. Similarly, in South Africa and Australia, settlers adapted and evolved property rights regimes to suit local conditions. Additionally, during the gold rushes there where shifts in the approaches to property rights in the various stages of the rushes, recognizing that there was a different need between "panners" and "miners" (i.e. those who panned for gold in a stream vs those who had to dig mine shafts). This ties in with the previous point about involving the people with the greatest stake in the process of developing the property rights, but it also means that there is a need to avoid being particularly dogmatic about the nature of property as applied to space resource activities and indeed any use of outer space. Space is a unique environment, drastically different from any humanity has encountered, it makes sense that property, as an institution, will have to adapt and evolve to deal with the differences.

The recent "demise" of the "pioneers" of asteroid based space resource activities, Deep Space Industries (acquired by Bradford

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[241] Outer Space Treaty, *supra* note 3, art VI.

[242] *Id.* art. II.

[243] HAGUE BUILDING BLOCKS, *supra* note 72, ¶¶ 4.2(b), 4.2(h), 4.3(c), 6.2, 7.1, 9(f), 10.1, 10.2(a), 11.1, 13(a), 13(b), 13(e), 13(f), 17(a), 17(b).

Engineering B.V) and Planetary Resources (acquired by Consen-Sys, Inc.), should be taken as an opportunity. There has been much activity, as has been discussed above, in various legislatures, at the United Nations, and through groups such as The Hague Space Resources Governance Working Group in the wake of the excitement generated by the initial announcements by Planetary Resources and Deep Space Industries several years ago, but the international community has been offered an opportunity to pause and reflect. There are still ventures pursuing space resource activities, although it now looks like the Moon is the likelier target for the first mining operations than any asteroid, but we should recognize that this is harder than and probably not as imminent as we, perhaps, once thought. While for many space enthusiasts this will be a disappointment, and indeed if humanity is truly to become a spacefaring civilization then it needs to develop an in-space economy and space resources, as well as dramatically reducing the cost of accessing space (at the very least by allowing in-orbit refueling, reducing the quantity of propellant needed to be hauled out of Earth's not insubstantial gravity well, as well as allowing for in-space manufacturing thus reducing the amount of stuff needed to be brought into space from Earth's surface), will be a foundation for that economy. However, as this project is an epochal endeavor, Elon Musk might talk about building cities on Mars within his lifetime[244] but the task of developing humanity into a spacefaring civilization will take centuries if not millennia. There is virtue in slowing down. There is no need to rush to produce legislation or regulation for space resource activities, we have the time and the scope to give the proper care and consideration to this issue, and we owe it to ourselves and future generations to do just that. Lawyers are often accused of being reactive to events but in this instance, we have been proactive, but we need to ensure that we are not too proactive, and indeed we must strive to avoid being pre-emptive. We have the time to get this right, let us at least make an effort to not repeat the mistakes of history. The work of The Hague International Space Resources Governance Working Group and the ongoing discussions at UNCOPUOS are great first steps, but there is more to be done.

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[244] Mike Wall, *SpaceX's Mars Colony Plan: How Elon Musk Plans to Build a Million-Person Martian City,* SPACE.COM (June 14, 2017), https://www.space.com/37200-read-elon-musk-spacex-mars-colony-plan.html.

THE HAGUE INTERNATIONAL SPACE RESOURCES GOVERNANCE WORKING GROUP: REFLECTIONS ON THE BUILDING BLOCKS FOR AN INTERNATIONAL FRAMEWORK

ABSTRACT

The Hague International Space Resources Governance Working Group (Working Group) is a cross-disciplinary, multinational group formed to specifically to address governance issues surrounding the maturing development and advancement of space resource activities. It is the intent of the Working Group to examine concepts that will enable and coordinate the use of space resources. The Working Group is hosted by an international consortium whose principal partner is the Institute of Air and Space Law at the Leiden Law School, Leiden University. The Center for Air and Space Law at the University of Mississippi School of Law, which sent an Associate Director to participate as an Observer at a number of the face-to-face meetings, applauds this important initiative and invited the Secretariat of the Working Group to share reflections with the Journal of Space Law, including insight into the purpose and procedures of the Working Group. The following Article was submitted by the Secretariat in response to this invitation. Part One of this Article explains the structure and functioning of the Working Group. Part Two provides details regarding the Working Group's main work product, namely draft building blocks for the development of an international framework on space resource activities to be offered as a basis for negotiations on an international agreement or non-legally binding instrument. Parts Three and Four conclude by highlighting the particular challenges that arise in respect of the governance of space resource activities, the contribution of the Working Group to the ongoing discussion and a way forward.

I. THE STRUCTURE AND OPERATION OF THE WORKING GROUP

The Hague International Space Resources Governance Working Group (Working Group or Group) was established in 2016 in the aftermath of The Hague Roundtable on the Governance of Space Mineral Resources organized by The Hague Institute for Global Justice.[1] At the Roundtable, political and legal experts agreed that a structured approach to the governance of space resources was essential given the recent emergence of activities focused on the exploitation of space mineral resources.[2]

The initial purpose of the Working Group was to assess the need for a framework on space resource activities. The prerequisite of compliance with international space law in conjunction with the need for certainty with regard to the legality of space resource activities confirmed the need for such framework. Subsequently, the Working Group proceeded to draw its ensuing objective: to lay the groundwork for a governance framework that could encourage States to engage in relevant discussions. In achieving these goals, the Working Group formulated a set of building blocks for the development of an international framework on space resource activities (Building Blocks). In addition, the Group began to identify strategies for the implementation of the Building Blocks, with the purpose of providing recommendations for governments and other entities that may initiate the advancement of a governance framework. The Working Group, which is managed by a Secretariat consisting of the Chair, two Vice-Chairs, the Executive Secretary and the Assistant Executive Secretary, was set up for a period of two years from 2016 to 2017. In 2017 the members of the Working Group decided to extend its mission for another two-year period from 2018 through 2019.

The Working Group is hosted by a consortium of partners (Consortium) from around the world including the Institute of Air and Space Law, Leiden Law School, Leiden University, the Catholic University of Santos, the Indonesian Centre for Air and Space Law,

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[1] *See The Hague Roundtable on the Governance of Space Mineral Resources*, THE HAGUE INSTITUTE FOR GLOBAL JUSTICE (Dec. 1, 2014), http://www.thehagueinstitute-forglobaljustice.org/events/the-hague-roundtable-on-the-governance-of-space-mineral-resources/.

[2] *Id.*

Padjajaran University, the Secure World Foundation, the University of Cape Town, the University of Luxembourg, the Nishimura Institute of Advanced Legal and the Ten to the Ninth Plus Foundation. Consortium partners communicate biannually to draw up plans for their collaboration in the development of the building blocks and the promotion of the results of the Working Group in their respective regions.

The operative body of the Working Group is comprised of members (Members) that are responsible for making decisions and expected to contribute to the ongoing discussions within the Working Group. The Working Group is also supported by an extensive network of observers (Observers) who provide input and receive any documentation disseminated within the Working Group. Members and Observers represent major stakeholders and institutions directly involved in space resource activities, ranging from governments, space agencies and industry, to academia, civil society and international organizations.[3] Whereas the number of Observers is unlimited, the maximum number of Members is set to thirty-five, increased from thirty during the first phase, in order to maintain a manageable and efficient platform of discussion.

Members and Observers are invited to participate in the face-to-face meetings, which take place every year in the spring and fall. The meetings of the first phase took place in Leiden and the meetings of the second phase are taking place in Leiden (spring) and in Luxembourg (fall). The purpose of the meetings is to produce the main deliverables of the Working Group, the Building Blocks and the recommendations for implementation strategies. In advance of each meeting, the Members and the participating Observers receive the agenda, along with meeting material consisting of the feedback received on the preceding discussions and any other relevant documentation. During the meetings the provisions of the Building Blocks are extensively analyzed and participants are encouraged to express their very varied opinions. Occasionally, the meetings include presentations from experts intended to inform and raise

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[3] For a full list of Members and Observers, see *The Hague International Space Resources Governance Working* Group, LEIDEN UNIVERSITY, https://www.universiteitleiden.nl/en/law/institute-of-public-law/institute-for-air-space-law/the-hague-space-resources-governance-working-group#first-face-to-face-meeting-of-the-second-phase,first-face-to-face-meeting (last visited Aug. 5, 2019).

awareness on issues relevant to space resource utilization, such as the scientific methods of research and extraction, the technical issues related to mining of space resources, the status of technological advancement, the economic aspects of space resource exploitation, as well as the ethical and environmental aspects of space resource activities. After each meeting the participants receive the notes and, when applicable, a revised version of the Building Blocks. The latter remains open for comments on behalf of Members and Observers until the following meeting, where new input is further discussed. External parties were also invited to comment on the Building Blocks prior to the Working Group's second face-to-face meeting in 2018.

In addition to the face-to-face meetings, the Working Group organizes outreach activities to promote and invite further input on the building blocks. The Working Group has also been present in several international events and fora, including the Legal Subcommittee of United Nations Committee on the Peaceful uses of Outer Space (UNCOPUOS) and the International Astronautical Congress.[4] At the 57th session of the Legal Subcommittee of UNCOPUOS, the Working Group organized a side-event to raise awareness on its activities among the UNCOPUOS delegates. The Dutch delegation also provides the UNCOPUOS with a regular a status report of the Working Group.[5] Moreover, the Members and Observers have presented the Building Blocks and the developments of the Working Groups to the events where they participate.

In parallel, the Secretariat informs the Members and Observers about progress with regard to the Building Blocks and about

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[4] *See e.g.,* T. Masson-Zwaan, R. Lefebre, G. Reibaldi & D. Stefoudi, *The Hague Space Resources Governance Working Group: Second Progress Report and the Way Forward*, 2017 INT'L INST. OF SPACE LAW 281 (2018); T. Masson-Zwaan, R. Lefebre, G. Reibaldi & M. Stewart, *The Hague Space Resources Governance Working Group*, 2016 OF THE INT'L INST. OF SPACE LAW 163 (2017).

[5] *See, e.g.,* The Hague Space Resources Governance Working Group, Information Provided by the Netherlands to the Legal Subcommittee of the U.N. Comm. on the Peaceful Uses of Outer Space, U.N. Doc. A/AC.105/C.2/2018/CRP.18 (Apr. 12, 2018); Comm. On the Peaceful Uses of Outer Space, Draft Rep. IV of the Legal Subcomm. on the Status and Application of the Five United Nations Treaties on Outer Space, ¶ 18 U.N. Doc. A/AC.105/C.2/L.301/Add.1 (Mar. 30, 2017); The Hague Space Resources Governance Working Group, Factsheet - Information Provided by the Netherlands to the Legal Subcomm. of the U.N. Comm. on the Peaceful Uses of Outer Space, U.N. Doc. A/AC.105/C.2/2016/CRP.17 (Apr. 5, 2016).

upcoming activities. The Secretariat also compiles a quarterly newsletter which is sent to a list of subscribers and provides updates to interested parties through the Working Group's social media.[6]

II. THE BUILDING BLOCKS

A. Introduction

The Building Blocks address various aspects of space resource activities and aim to serve as a baseline for further discussion on a potential future framework on space resource activities. They comprise 19 articles which cover substantive and non-substantive matters, including access to and rights over space resources, sharing of benefits from space resource activities and the establishment of safety zones around areas of extraction. Their content has been extensively discussed during the face-to-face meetings of the Working Group. The evolving versions of the draft Building Blocks have been published online in updated versions. The preliminary results of the first phase of the Working Group were made available online in September 2017.[7]

The Working Group found that the most challenging legal questions with regard to space resource utilization are twofold: the first issue involves the prohibition of national appropriation of "[o]uter space, including the Moon and other celestial bodies," set forth in Article II of the Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies (Outer Space Treaty);[8] and the second issue centers on the sharing of benefits resulting from

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[6] *The Hague International Space Resources Working Group*, FACEBOOK,https://www.facebook.com/TheHagueSpaceResourcesGovernanceWG (last visited Aug. 5, 2019); Space Resources Working Group (@SpaceResourceWG), TWITTER, https://twitter.com/SpaceResourceWG (last visited Aug. 5, 2019).

[7] THE DRAFT BUILDING BLOCKS FOR THE DEVELOPMENT OF AN INTERNATIONAL FRAMEWORK ON SPACE RESOURCE ACTIVITIES (2017), https://www.universiteitleiden.nl/binaries/content/assets/rechtsgeleerdheid/instituut-voor-publiekrecht/lucht—en-ruimterecht/space-resources/draft-building-blocks.pdf [hereinafter BUILDING BLOCKS].

[8] Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space Including the Moon and Other Celestial Bodies art. II, Jan. 27, 1967, 18 U.S.T. 2410, 610 U.N.T.S. 205 [hereinafter Outer Space Treaty].

the undertaking of space resource activities, a concept that is subject to interpretation based on Article I of the Outer Space Treaty which states that the exploration and use of space "shall be the province of all [hu]mankind."[9]. The Building Blocks take into account the principles of international space law and attempt to interpret them in the context of space resource activities, as well as suggest a balanced approach among the legal, technical, scientific and commercial aspects of space resource activities.

The following paragraphs elaborate on the provisions of the Building Blocks and other issues that surfaced during the exchange of opinions among the participants. The Building Blocks reflect the topics that have been suggested by the Working Group for consideration toward the potential development of a framework of space resource activities.

B. General Provisions and Definitions

The Building Blocks are structured upon the concept of international space law, which prescribes rights and obligations for the States parties to the international treaties that govern space activities, including the Outer Space Treaty, the Rescue Agreement,[10] the Liability Convention,[11] the Registration Convention,[12] and the Moon Agreement[13] (together, Space Treaties). Therefore, they aim to provide recommendations to the States and intergovernmental organizations that may negotiate a framework for the governance of space resource activities.[14]

Even though they do not directly address private actors involved in such activities, their interests are taken into consideration in the drafting of the provisions. The balance between legal and commercial implications is reflected in the proposed objective of the framework, namely the creation of an enabling environment for the

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[9] *Id.* art. I.

[10] Agreement on the Rescue of Astronauts, the Return of Astronauts and the Return of Objects Launched into Outer Space, Apr. 22, 1968, 672 U.N.T.S. 119.

[11] Convention on International Liability for Damage Caused by Space Objects, Mar. 29, 1972, 961 U.N.T.S. 187 [hereinafter Liability Convention].

[12] Convention on Registration of Objects Launched into Outer Space, June 6, 1975, 1023 U.N.T.S. 15 [hereinafter Registration Convention].

[13] The Agreement Governing the Activities of States on the Moon and Other Celestial Bodies, Dec. 5, 1979, 1363 U.N.T.S. 3 [hereinafter Moon Agreement].

[14] BUILDING BLOCKS, *supra* note 7, ¶ 3.

development of space resource activities in the interest and for the benefit of all countries and humankind.[15] Toward this end, the framework is designed to: 1) comply with the principles of international space law; 2) recommend governance mechanisms to States and intergovernmental organizations; and 3) identify best practices in the field of space resource activities.

In general, consistency with the provisions of the space treaties is stressed throughout the text of the Building Blocks. In particular, Building Block 4 refers to principles similar to those of the Outer Space Treaty. For example, the Building Blocks adhere to the language of the Outer Space Treaty in stating that: "space resources shall be used exclusively for peaceful purposes;"[16] space resource activities "shall be "carried out for the benefit and in the interests of all countries and humankind irrespective of their degree of economic and scientific development;"[17] "space resource activities shall not harmfully interfere with other on-going space activities, including other space resource activities;"[18] and "international cooperation in space resource activities shall be conducted in accordance with international law."[19]

Building Block 4 also recognizes that the framework should provide legal certainty and correspond to contemporary technological advancement, in accordance with the principle of adaptive governance.[20] This point was added to underline the impact of appropriate regulation in the development of space resource activities.

Building Block 2 provides definitions of the terms that are used throughout the text. Except from the definition of space object which mirrors Article I of the Liability Convention and of the Registration Convention,[21] the rest of the terms are specific to the characteristics of space resource activities and take into account their particular legal, technical and commercial aspects.

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[15] *Id.* ¶ 1.

[16] *Id.* ¶ 4.3(a); *see also* Outer Space Treaty, *supra* note 8, art. IV.

[17] BUILDING BLOCKS, *supra* note 7, ¶ 4.3(b); *see also* Outer Space Treaty, *supra* note 8, art. I.

[18] BUILDING BLOCKS, *supra* note 7, ¶ 4.3(c); *see also* Outer Space Treaty, *supra* note 8, art. IX.

[19] BUILDING BLOCKS, *supra* note 7, ¶ 4.3(d); *see also* Outer Space Treaty, *supra* note 8, art. III.

[20] BUILDING BLOCKS, *supra* note 7, ¶ 4.2.

[21] *See* BUILDING BLOCKS, *supra* note 7, ¶ 2; Liability Convention, *supra* note 11, art. 1; Registration Convention, *supra* note 12, art 1.

To start, space resources are defined as the abiotic resources that are extractable and/or recoverable and are found *in situ* in outer space.[22] It is clarified that space resources include mineral and volatile materials, such as water, but exclude elements that are regulated by different regimes, as, for example, satellite orbits and radio spectrum, or do not directly associate to the measurable nature of resources, such as the energy from the sun.[23] On the same note, utilization of space resources is defined as the recovery of resources and the extraction of raw minerals or volatile materials therefrom.[24] This definition was shaped in this manner to differentiate the utilization of space resources from the means of appropriation of outer space as described in Article II of the Outer Space Treaty. The definition distinctly does not extend to the secondary uses of the extracted material, since the Building Blocks do not intend to address issues that do not have direct implications with the legal status of space resources.[25]

Space resource activities on the other hand are defined in a broader manner, in order to ensure that the framework would cover the entire spectrum of the activities connected to the extraction and recovery of space resources. The term includes the search for and utilization of space resources, as well as the construction and operation of associated systems for extraction, recovery, processing and transportation of those resources.[26] Furthermore, Building Block 2 defines space products as products made in outer space "wholly or partially from space resources,"[27] and operators as the "governmental, intergovernmental or non-governmental entities"[28] that conduct space resource activities, so as to capture all the stakeholders currently and potentially involved. The Working Group has extensively discussed the aforementioned definitions, in order to guarantee their accuracy in terms of the existing legal framework and current technological developments.

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[22] BUILDING BLOCKS, *supra* note 7, ¶ 2.1.
[23] *Id.* n.2.
[24] *Id.* ¶ 2.2.
[25] *Id.* n.3.
[26] *Id.* ¶ 2.3.
[27] *Id.* ¶ 2.5.
[28] *Id.* ¶ 2. 6.

C. Space Law Principles in the Context of Space Resource Activities

The Building Blocks suggest that the Space Treaties be interpreted in a manner that reflects the characteristic elements of space resource activities, in order to ensure compliance of the latter with the former and with existing principles of international law.

That said, the Building Blocks extend the substantive principle where possible. For example, under Article VI of the Outer Space Treaty, States "bear international responsibility for national activities in outer space . . . whether such activities are carried on by governmental or non-governmental entities."[29] The Building Blocks extend this responsibility to all "space resource activities authorized by them,"[30] in order to eliminate the possibility of unauthorized space resource activities. Moreover, jurisdiction and control over space objects, as set forth in Article VIII of the Outer Space Treaty[31] is expanded by the Building Blocks to include space-made products, so as to ensure that they are also properly overseen.[32] Similarly, Building Block 15 calls for the applicability of the liability provisions of the Outer Space Treaty and the Liability Convention to damages caused by space resources activities, as well as for the encouragement of space resource operators to arrange compensation for damages in other means.[33]

The Building Blocks also include specific provisions to assure the avoidance of harmful interference in the space activities of others. Building Block 9 proposes the adoption of a "precautionary approach" in the conduct of space resource activities so as to avoid and mitigate potentially harmful impact, including, among others, to the safety of persons, to property and to the environment of the Earth and of outer space.[34] It also makes particular mention of the need for a precautionary approach to safeguard sites in outer space

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[29] Outer Space Treaty, *supra* note 8, art. VI.
[30] BUILDING BLOCKS, *supra* note 7, ¶ 5.1.
[31] Outer Space Treaty, *supra* note 8, art. VIII.
[32] BUILDING BLOCKS, *supra* note 7, ¶ 5.3.
[33] *Id.* ¶ 15.
[34] *Id.* ¶ 9(a)-(e).

that are "designated and internationally endorsed outer space natural or cultural heritage sites,"[35] and to established planetary protection policies.[36] Additionally, Building Block 11 calls for States and intergovernmental organizations to monitor and redress harmful impact from the space resource activities they authorize and supervise, as well as to adopt mechanisms for response and adjustment or termination of space resource activities in case of foreseeable harmful impact.[37] Along the same lines, Building Block 8, again similar to the Outer Space Treaty, stresses that the framework should require States and organizations to pay due regard to the interests of all countries in the conduct of the space resource activities they authorize and supervise.[38]

Building Block 13 calls for the registration, pursuant to the Registration Convention of the space objects launched to carry out space resource activities and for the notification of frequency allocation for such activities, in accordance with the regime of the International Telecommunication Union (ITU).[39] In order to promote the sharing of information described in Article XI of the Outer Space Treaty, Building Block 13 also suggests that States and intergovernmental organizations share information related to the space resource activities they authorize and supervise. In particular, it recommends the notification about the purpose, technical characteristics and duration of the activities, the nature and location of space resources, associated logistic activities, the result and impact of the activities, as well as their termination.[40]

The issue of the sharing of benefits was extensively discussed prior to the drafting of Building Block 12. The Outer Space Treaty states that the exploration and use of outer space shall be carried out for the benefit and in the interest of all countries and shall be the province of humankind.[41] Lacking further details as to the

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[35] *Id.* ¶ 9(g).

[36] *Id.* ¶ 9(d).

[37] *Id.* ¶ 11.

[38] BUILDING BLOCKS, *supra* note 7, ¶ 8; *see also* Outer Space Treaty, *supra* note 8, art. IX.

[39] BUILDING BLOCKS, *supra* note 7, ¶ 13.

[40] BUILDING BLOCKS, *supra* note 7, ¶ 13(e); *see also* Outer Space Treaty, *supra* note 8, art. XI.

[41] Outer Space Treaty, *supra* note 8, art. I.

methods of sharing and to the definition of the province of humankind, the Treaty allows room for various interpretations. Some initial concerns, voiced primarily on behalf of industry representatives, were related to potential legal obligations to share the proceeds garnered from their space resource activities, which might be a disproportionate or disincentivizing measure given the significant investments made especially by pioneer operators. In an attempt to associate the benefit-sharing principle to the commercial practice in the field of space resources, the Working Group explored options for the fulfilment of the purpose of the principle. Toward this end, it recommended that the framework require that States and intergovernmental organizations promote the participation to space resource activities by all countries, particularly developing countries.[42] The Working Group also proposed specific examples of benefits that may be shared though the promotion of participation, including the development of space science and technology, the development of capabilities in interested States, cooperation in education and training, the exchange of and access to information and expertise, the collaboration in joint ventures, as well as the establishment of an international fund.[43] Building Block 12 does not provide for an exclusive list of methods for benefit-sharing, nor does it support compulsory monetary benefit-sharing. On the contrary, it intends to illustrate feasible measures for States and intergovernmental organizations, as well as the private sector.

The Building Blocks also encompass some of the non-substantive provisions of the Space Treaties. Like the Outer Space Treaty, the Building Blocks indicated that the framework should require the provision of assistance in case of distress;[44] as well as reciprocal visits of stations, equipment, installations and vehicles relating to space resource activities as provided.[45] Moreover, in the spirit of cooperation among States encouraged by the Space Treaties, Building Block 18 suggests the peaceful settlement of disputes arising from space resource activities through, consultation and arbitration.[46]

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[42] BUILDING BLOCKS, *supra* note 7, ¶ 12.1.

[43] *Id.*

[44] BUILDING BLOCKS, *supra* note 7, ¶ 14; *see also* Outer Space Treaty, *supra* note 8, art. V.

[45] BUILDING BLOCKS, *supra* note 7, ¶ 16; *see also* Outer Space Treaty, *supra* note 8, art. XII.

[46] BUILDING BLOCKS, *supra* note 7, ¶ 18.

D. Additional Provisions

Outside adapting the concepts of the Space Treaties into the context of space resource activities, the Building Blocks propose additional provisions with regard to the rights of space resource operators over areas and the extracted resources and the establishment of safety zones. Given the technical and financial burden that the operators undertake, as well as the risky nature of activities in outer space, these provisions were thoroughly assessed in order to ensure their accordance with the Space Treaties and their proportionality compared to the challenges to which they correspond.

Building Block 6 seeks to combine the principle of free access to all areas of outer space with the need of space resource operators for priority to secure their interests, given the time and capital consumed – at least initially – by the activities they undertake. It thus calls for the framework to enable the attribution of priority rights to operators for the *in situ* search and recovery of space resources and to provide for their recognition on an international level.[47] In order to avoid conflict with the freedom of access, it specifies that such rights should be provided for a maximum period and a maximum area, and be granted based on the circumstances of each individual activity.[48] This way, the freedom of exploration and use is narrowed and restricted only as distinctly needed. Building Block 6 also suggests the registration of priority rights in an international registry,[49] so as to further ensure conformity with the Outer Space Treaty and increase transparency in the conduct of space resource activities.

Similarly, Building Block 7 addresses rights over space resources and the products derived therefrom. It explicitly mentions that the framework should provide for such rights in a manner that would not "contravene the principle of non-appropriation"[50] of Article II of the Outer Space Treaty, which prohibits the appropriation of outer space "by means of sovereignty, occupation or any other means."[51] Even though the legality of the use of space resources vis-

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[47] *Id.* ¶ 6.2.

[48] *Id.*

[49] *Id.* Building Block 13 describes the details of registration of such rights. *See id.* ¶ 13.

[50] BUILDING BLOCKS, *supra* note 7, ¶ 7.

[51] Outer Space Treaty, *supra* note 8, art. II.

à-vis the prohibition of appropriation has been contested, the Working Group supports the view that the extraction of resources does not fall under the scope of the prohibition. Resource rights are of cardinal importance for the performance of space resource activities in a lawful manner, thus Building Block 7 reiterates the need for the framework to recognize the legitimacy of the acquisition of space resources, in accordance with the non-appropriation principle.

Finally, Building Block 10 aims to harmonize the safety requirements in the conduct of space resource activities with the aforementioned freedom of access and non-appropriation principle. In general, it recommends that authorization and supervision of space resource activities by States and intergovernmental organizations be based on prior review of their safety.[52] In particular, it further proposes the establishment of safety zones or other area-based safety measures for an identified radius around space resource activities.[53] The latter should be extended as necessary to assure the safety of the activities and to avoid harmful interference with it. Given that the purpose of such measure is strictly the safe conduct of space resource activities, Building Block 10 underlines as well the accordance with the non-appropriation principle. Similar to the priority rights concept encompassed in Building Block 6, a safety zone is foreseen for a limited period of time and following timely public notice to avoid impeding the conduct of other planned missions and limiting the freedom of access.[54]

E. Technical and Socio-economic Aspects of Space Resource Utilization

At the beginning of its second phase, the Working Group introduced two panels to provide specific advice with regard to the text of the Building Blocks, particularly in terms of their feasibility in the current framework of space resource activities. Panels were created to focus on technical aspects (Technical Panel) and a socio-economic aspect (Socio-Economic Panel and, together with the Technical Panel, Panels) of space resource utilization in order to foster

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[52] BUILDING BLOCKS, *supra* note 7, ¶ 10.1.
[53] *Id.,* ¶ 10.3.
[54] *Id.*

dialog and cooperation among stakeholders and provide insight on these aspects to the Working Group. The Panels consist of experts in different fields, including among others policy, engineering, physics, social sciences, economics, environmental studies and security and hold regular teleconferences. They are managed by a Panel Chair and report to the Working Group during the face-to-face meetings.

The Technical Panel has divided its work into three sub-categories, namely the allocation of frequencies for lunar and deep space missions, the identification of standards for the establishment of safety zones and the technical extensions of the Building Block provisions. The Panel participants work in teams that focus on one of the said topics and examine hypothetical scenarios to validate their positions. In terms of general feedback, the Technical Panel has highlighted the significance of adaptive governance and suggested the periodic review of the terms of the governance framework in order to sustain an innovation-friendly environment. Furthermore, the Panel has supported the involvement of private actors in the identification of best practices in the field of space resources, given their hands-on experience on this subject. The Technical Panel has also recommended that the definitions of space objects and space-made products, as presented in the context of responsibility and avoidance of harmful interference, account for the residual material from a terminated activity, as well as the returned samples of resources.

With respect to frequency allocation, the Technical Panel has acknowledged the exclusive mandate of the ITU to allocate and register radio frequencies. The ITU offers a well-established regime for the coordination of frequencies around the Earth's orbits for public and private missions. However, in the absence of provisions for commercial missions, activities around the Moon and in other parts of outer space are currently registered as research or operational missions.[55] To address this shortcoming, the Panel proposes the development of a dedicated framework for lunar and deep space frequency allocation for commercial purposes, complementing the existing ITU regime.

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[55] *See* ITU Radio Regulations 2016, Article 22.22, *available at* https://www.itu.int/pub/R-REG-RR.

Bearing in mind the lessons learnt from the allocation of frequencies in the geostationary orbit, the Panel suggests the pursuit of such initiative through the official ITU procedures, in order to mitigate risks of harmful interference from space resource activities. Moreover, the Panel has elaborated on the provisions of Building Block 10 pertaining to the conditions and purpose of the establishment of safety zones around space resource activities. In particular, it has pointed out the difficulty of introducing a uniform system for the designation of safety zones, given the varying types of activities and environmental conditions in outer space. Therefore, in the assessment of the requirements for the establishment of safety zones, the drafters of the governance framework should be driven by the goal of avoiding harmful interference and ensuring the safety of the personnel and equipment involved in space resource activities. To this end, the Technical Panel suggests the sharing of information among operators for the coordination of traffic from different missions and the introduction of measure to guarantee safe proximity among the missions. Moreover, with respect to the conditions for setting up safety zones, the framework should also include details about their maintenance and termination, taking into account the characteristics of each envisioned area of extraction.

The Socio-Economic Panel has identified five areas of focus influenced by existing social and financial concepts, including examples from other sectors. First, the Socio-Economic Panel examined the relevance of social licenses for the operation of space resource activities. This notion refers to the assessment of activities based on their level of acceptance on behalf of the community in which they are conducted and associates to concerns over the sustainability of space resource activities and their potential impact on other markets and industries. Second, the Socio-Economic Panel reviewed the benefits of responsible investment and its application in the terrestrial mining sector. In particular, it considered ways in which corporations involved in space resource activities and the investors who support such activities can implement methods of appropriate governance in all stages of their involvement. Third, the lessons learned from the sharing of data in the Earth observation and other space fields are promoted as examples of cooperation among the operators of space resource activities. Fourth, the Socio-

Economic Panel attempted to define capacity-building mechanisms in an effort to broaden the scope of the benefits deriving from space resource activities. For this purpose, it explored existing capacity-building programs related to space resources and other space activities. Fifth, the Panel looked into models for cooperation among States and operators in the advancement of space resource activities and the value accrued from such initiatives, aiming to define the types and objectives of potential collaboration.

The aforementioned concepts introduced by the Panels are currently being examined in conjunction with the provisions of the Building Blocks, in an effort to suggest improvements to their text and appropriate methods for their implementation.

F. Perspectives on the Progress of the Building Blocks

As previously stated, the purpose of the Building Blocks is to identify issues that should be addressed in a future framework on the governance of space resource activities. In principle, they are not meant to spell out specific regulatory provisions, but to serve as baseline for further discussion. Therefore, their scope is intentionally general so as to enjoy wide acceptance and to remain open for further specification by the States or other parties that may eventually engage in negotiations. The 19 provisions of the Building Blocks are not exhaustive in terms of matters related to space resource activities or binding to any extent. The findings of the Working Group represent the insightful opinions and expert knowledge of its participants, who have recognized certain topics as essential to mention, taking into account current and future business perspectives, technological developments and legal implications.

Particular attention has been paid to the language of the Building Blocks so as to coincide with the terms used in the Space Treaties and to refrain from inconsistency among the provisions. Nevertheless, certain notions have been intentionally left undefined or vague, in order to be specified by the parties that will follow through with negotiations. Some of the omitted elements are specified in Building Block 17 on institutional arrangements, which proposes methods for the framework to address the designation of heritage and scientific sites, the establishment of a registry for resource rights and activities and the identification of standards and